Decentralised federated k-means clustering using GTV Mimimisation.

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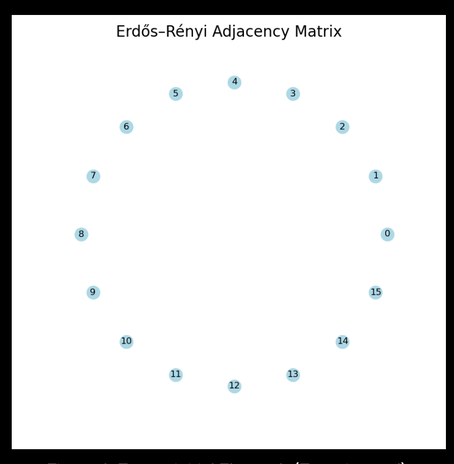

Figure 1: Empty initial FL graph (Experiment 1).

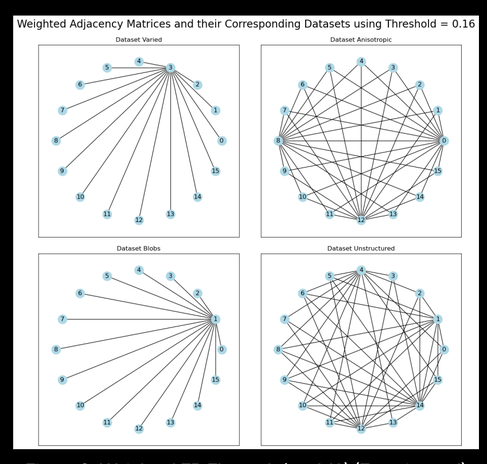

Figure 2: Weighted ER FL graph (t = 0.16) (Experiment 1)

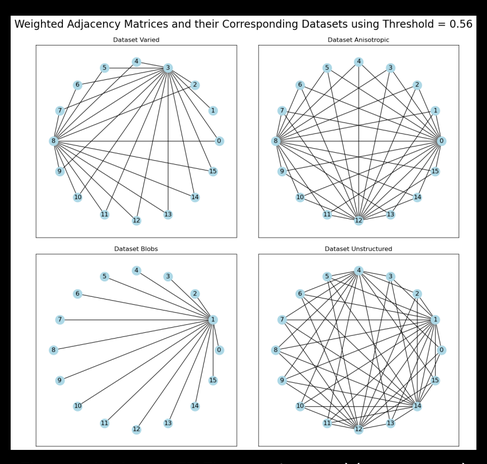

Figure 3: Weighted ER FL graph (t = 0.56) (Experiment 1)

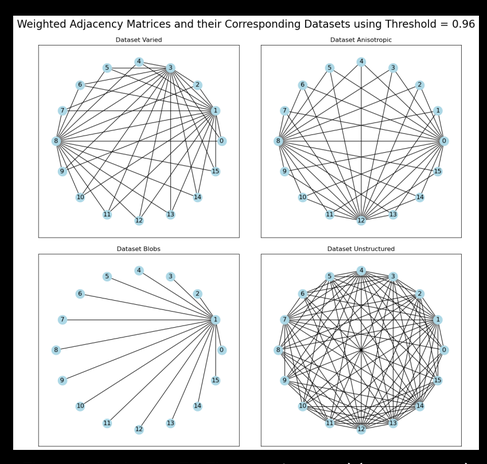

Figure 4: Weighted ER FL graph (t = 0.96) (Experiment 1)

What is GTV Minimisation (GTVMin)?

The Generalized Total Variation (GTV) of a local machine learning model is defined as [1]

$$\mathrm{GTVMin}(\{h^{(i)}\}) = \sum_{\{i,i'\}\in E} A_{i,i'} d(h^{(i)}, h^{(i')}),$$

where $A_{i,i'} > 0$ is the weight of edge $\{i, i'\}$.

GTVMinimisation (GTVMin) defines a objective function for the machine learning algorithm such that [1]

$$\hat{h} \in \mathrm{argmin}_{h\in H}\ L_b(h|D) + \alpha \mathrm{GTV}(h^{(i)}, h^{(i')}),$$

where $L_b(h|D)$ is the empirical risk on the training data D and $a > 0$ controls the strength of the GTV regularisation.

When performing the federated variant of k-means clustering, communication of model parameters was controlled using an Erdős–Rényi adjacency matrix. This provides an alternative method of communication between nodes, instead of the standard set-up of a central server communicating to a satellite of nodes.

Initially, the adjacency matrix took binary values to represent connections between nodes i and j such that nodes i, i' are neighbours if $A[i, j]|_{i,i'} = 1$.

The next step was to add weights to the edges of the FL graph proportionate to the similarity of each client's dataset. This was achieved by calculating the pairwise differences between the log density (the logarithm of the probability density function) of each client.

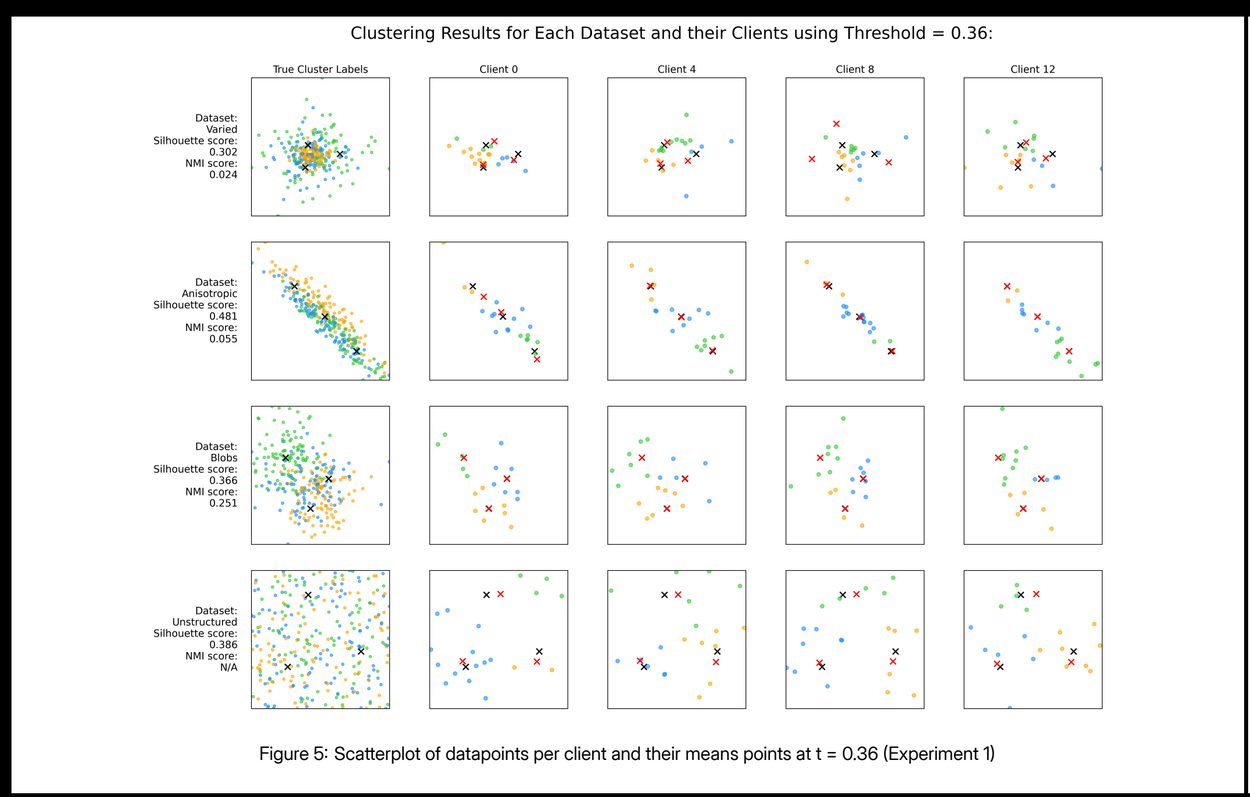

Figure 5: Scatterplot of datapoints per client and their means points at t = 0.36 (Experiment 1)

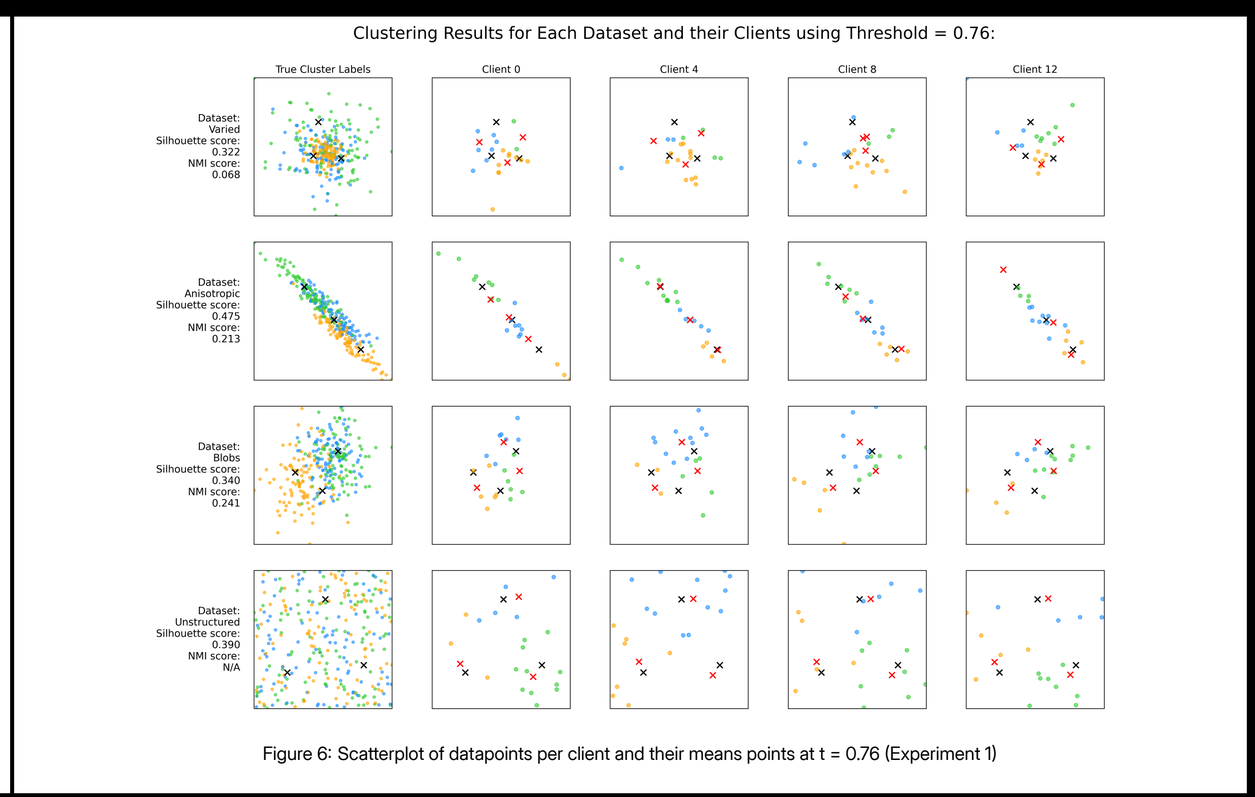

Figure 6: Scatterplot of datapoints per client and their means points at t = 0.76 (Experiment 1)

These edge weights multiplied α to control the strength of the GTV term in the minimisation objective function. In this way, clients with more similar datasets would share more information between each other in the FL network. Conversely, it acted as a penalty term for clients with more dissimilar datasets.

When building the FL graph for each shape of dataset (as seen in Figures 5 and 6), clients with more dissimilar datasets were connected first, with the threshold t-value increasing with each iteration.

Figure 7 shows a decrease in centroid consensus as the t-value increases. The trend lines of Figure 7 show the positive correlation between the threshold t-value and the divergence of centroids across clients.

The bar charts opposite show that unstructured datasets and datasets with higher standard deviation reliably produced greater variance in centroid difference across clients, whilst anisotropic datasets produced the least.

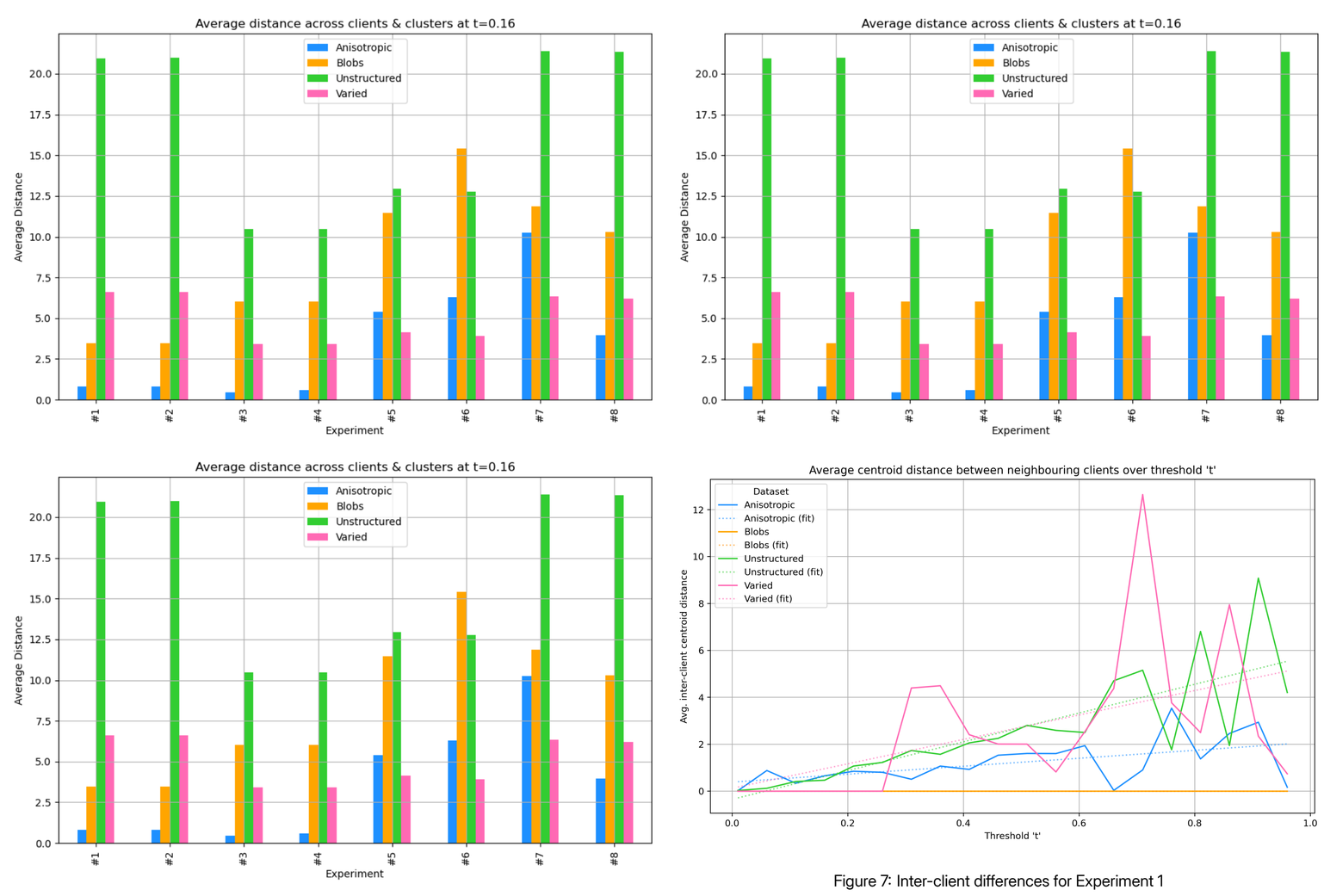

Figure 7: Inter-client differences for Experiment 1

References:

[1] Alexander Jung, Konstantina Olioumtsevits and Juliette Gronier, The Aalto Dictionary of Machine Learning, 2025, https://aaltodictionaryofml.github.io/